

MAT 504 — Computer Modelling in Materials Science

Week 1 Hands-on

From Interatomic Interactions to Mechanical Stiffness

Objective. In the lecture, we considered an engineering problem in which a continuum model requires a material property such as Young’s modulus. In this exercise, we will work in the opposite direction: starting from a simple model of atomic interactions, we will examine how a mechanical response can emerge from the microscopic model.

This is intentionally a simplified system. The aim is not to construct a realistic material model, but to understand the sequence

interaction model $\longrightarrow$ energy $\longrightarrow$ force $\longrightarrow$ mechanical response.

By the end of the session, you should be able to:

- interpret a pair potential physically;
- relate force to the derivative of potential energy;
- identify an equilibrium separation;
- relate the curvature of an energy minimum to local stiffness;
- obtain a force–extension response for a simple atomic chain;
- explain what assumptions are contained in the model.

Computational setup

We will use Python with

```
import numpy as np
import matplotlib.pyplot as plt
```

A starter Jupyter notebook will be provided. Complete the indicated cells and answer the short questions in the notebook.

Unless otherwise stated, use **reduced units**:

$$\epsilon = 1, \quad \sigma = 1.$$

Do not attach physical units to numerical values unless explicitly requested.

Part 1 — A simple interaction model

We will use the Lennard–Jones pair potential

$$U(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right].$$

Here, r is the distance between two particles. The two terms represent competing physical effects:

- short-range repulsion;
- longer-range attraction.

Task 1.1 — Implement the potential

Complete the function

```
def lj_energy(r, epsilon=1.0, sigma=1.0):
    # complete this function
```

Evaluate $U(r)$ over a suitable range of separations, for example

```
r = np.linspace(0.95, 3.0, 1000)
```

and plot $U(r)$.

Your plot should include:

- labeled axes;
- the line $U = 0$;
- the location of the energy minimum.

From your plot, estimate the equilibrium separation r_0 .
Compare your numerical result with

$$r_0 = 2^{1/6}\sigma.$$

What is physically special about this separation?

Part 2 — From energy to force

For a pair interaction, the radial force is

$$F(r) = -\frac{dU}{dr}.$$

For the Lennard–Jones potential,

$$F(r) = \frac{24\epsilon}{r} \left[2 \left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right].$$

Task 2.1 — Implement and plot the force

Complete

```
def lj_force(r, epsilon=1.0, sigma=1.0):
    # complete this function
```

Plot $F(r)$ over a range that clearly shows the behavior close to r_0 .

Mark:

- the point $r = r_0$;
- the region $r < r_0$;
- the region $r > r_0$.

Explain the sign of the force in each region:

1. $r < r_0$
2. $r = r_0$
3. $r > r_0$

Relate your answer to the shape of $U(r)$.

Task 2.2 — Check the energy–force relationship numerically

Use a numerical derivative:

```
F_numeric = -np.gradient(U, r)
```

Compare this with the analytical force calculated above. You do not need perfect agreement near the endpoints.

Why does the force vanish at the minimum of the potential energy?

Part 3 — Microscopic stiffness

Close to equilibrium, a smooth interaction potential can be approximated by a harmonic form:

$$U(r) \approx U(r_0) + \frac{1}{2}k_{\text{pair}}(r - r_0)^2,$$

where

$$k_{\text{pair}} = \left. \frac{d^2U}{dr^2} \right|_{r=r_0}.$$

The curvature of the potential therefore provides a measure of the local stiffness of the interaction.

Task 3.1 — Estimate the curvature numerically

Using a small displacement h , calculate

$$k_{\text{pair}} \approx \frac{U(r_0 + h) - 2U(r_0) + U(r_0 - h)}{h^2}.$$

For example, begin with

```
h = 1e-4
```

Suppose we keep σ fixed but increase ϵ .

Before running the calculation, predict what will happen to k_{pair} .

Then calculate k_{pair} for

$$\epsilon = 0.5, 1.0, 2.0.$$

Was your prediction correct?

Part 4 — From a pair interaction to a simple material

We now construct a highly simplified one-dimensional material.

Consider N identical atoms arranged in a line. Only nearest-neighbor atoms interact.

At equilibrium,

$$r = r_0,$$

and therefore the equilibrium length of the chain is

$$L_0 = (N - 1)r_0.$$

Use

$$N = 20.$$

If the chain is uniformly stretched to a new length L ,

$$r = \frac{L}{N - 1}.$$

The total interaction energy is

$$U_{\text{tot}}(L) = (N - 1)U(r).$$

Task 4.1 — Stretch the chain computationally

Define a small engineering strain

$$\varepsilon_{\text{strain}} = \frac{L - L_0}{L_0}.$$

Generate strains approximately between

$$-0.03 \leq \varepsilon_{\text{strain}} \leq 0.03.$$

For each strain:

1. calculate L ;
2. calculate the new nearest-neighbor separation r ;
3. calculate $U_{\text{tot}}(L)$.

Plot total energy versus strain.

At approximately what strain is the minimum energy found?
Should this result have been expected?

Task 4.2 — Obtain the force–extension response

The external force required to stretch the chain can be obtained from

$$P(L) = \frac{dU_{\text{tot}}}{dL}.$$

Use a numerical derivative:

```
P = np.gradient(U_total, L)
```

Define

$$\Delta L = L - L_0$$

and plot

$$P \quad \text{vs.} \quad \Delta L.$$

Focus first on the small-deformation region around $\Delta L = 0$.

Does the force–extension curve appear approximately linear close to equilibrium?
What familiar macroscopic relation does this resemble?

Part 5 — Connecting microscopic and macroscopic stiffness

For sufficiently small deformation,

$$P \approx K_{\text{chain}} \Delta L.$$

Estimate K_{chain} by fitting a straight line to the region close to equilibrium.

For example:

```
mask = np.abs(strain) <= 0.01

coeff = np.polyfit(
    delta_L[mask],
    P[mask],
    1
)

K_chain = coeff[0]
```

For $N - 1$ identical interactions connected in series, we expect approximately

$$K_{\text{chain}} = \frac{k_{\text{pair}}}{N - 1}.$$

Task 5.1

Calculate both

$$K_{\text{chain}}^{\text{fit}}$$

from the force–extension curve and

$$K_{\text{chain}}^{\text{predicted}} = \frac{k_{\text{pair}}}{N - 1}.$$

Compare the two values.

Explain, in your own words, why a material containing more identical bonds in series becomes easier to extend as a whole even though the individual interaction has not changed.

Part 6 — What did we actually model?

Our chain is **not a realistic model of a bulk solid**.

Discuss with the person next to you and identify at least **three important physical features that are absent from this model**.

Possible categories to think about include:

- dimensionality;
- atomic structure;
- temperature;
- defects;
- interactions beyond nearest neighbors;
- chemical bonding;
- surfaces;
- transverse deformation.

In the continuum beam problem from the lecture, Young's modulus E was an input to the model.

What has changed in the microscopic description used here?

Write **2–3 sentences** explaining the connection between

interatomic potential $\rightarrow$ microscopic stiffness $\rightarrow$ macroscopic mechanical response.

Before you leave

Make sure your notebook contains:

1. the Lennard–Jones energy curve;
2. the corresponding force curve;
3. your estimate of r_0 ;
4. your estimate of k_{pair} ;
5. the chain energy versus strain;
6. the force versus extension curve;
7. the fitted chain stiffness;
8. your short written answers to Questions 1–7 and the final question.

Keep this notebook. The first homework will build directly on the ideas and code developed today.

Suggested in-class timing

Time	Activity
0–5 min	Introduction and notebook setup
5–15 min	Part 1: potential energy
15–25 min	Part 2: forces
25–35 min	Part 3: microscopic stiffness
35–50 min	Parts 4–5: atomic chain and force–extension response
50–60 min	Interpretation, assumptions, and class discussion